

UX RESEARCH & HEURISTIC EVALUATION

Why booking a train ticket is harder than it needs to be

A structured usability evaluation of the IRCTC train ticket booking flow, applying Nielsen's ten usability heuristics and six established UX laws to India's highest-volume consumer transaction.

Product: IRCTC web portal & Rail Connect (NGeT e-ticketing platform)

Flow evaluated: Search → Select train → Traveller details → Verification → Payment

Method: Expert heuristic evaluation, severity-rated 0-4

Issues identified: 9 (three rated severity 4)

Evaluation date: September 2026

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Date: **September 2026**

Executive summary

IRCTC is the only official channel for booking reserved travel on Indian Railways. On 7 September 2026 it recorded its highest-ever single-day volume of 20,06,353 online tickets, and roughly 89% of all reserved rail tickets are now booked digitally. At that scale, a usability defect is not an inconvenience — it is a tax levied on tens of millions of people who have no alternative supplier.

This evaluation walked the complete booking flow as two different users — a first-time traveller and an experienced Tatkal booker — and assessed each screen against Nielsen's ten heuristics and six UX laws. Nine distinct usability issues were identified and severity-rated. Three are rated severity 4, meaning they should be fixed before any further feature work: availability information is presented in railway back-office notation that untrained users cannot decode; the booking session expires silently and destroys entered data; and the total payable changes after the user has already committed to a train.

The nine issues share a single root cause. The interface exposes the railway's internal operating model — quotas, waiting-list classes, session locks, payment-gateway fee structures — rather than translating it into the traveller's model, which contains only three questions: *will I get a seat, when do I leave, and what does it cost?* Most of the recommended fixes are therefore translation problems rather than engineering problems, and four of them are achievable through copy, validation and default changes alone.

Scope and honesty note

This report evaluates the production NGeT booking flow as of September 2026. IRCTC released a redesigned interface in beta on 15 July 2026; several recommendations here move in the same direction. Findings are therefore expressed as principles rather than as pixel-level redlines, so that they remain valid against either version of the interface.

The interface screens reproduced in this report are **faithful recreations built for annotation**, not live screen captures. They reproduce the structure, information density, terminology and control layout of the production flow so that specific elements can be circled and numbered. Where a reader needs pixel-exact evidence, the live site should be consulted directly.

1. Objective and scope

The objective of this study is to apply usability heuristics and UX laws to a real, live consumer interface; to identify and justify a minimum of five usability issues; to map each to the principles it violates; and to propose improvements that a product team could act on.

In scope

- The reserved-ticket booking flow from landing page through to payment initiation.
- Desktop web as the primary surface, with mobile-web implications noted where target size or thumb reach is material.
- Both the General quota flow (unhurried) and the Tatkal flow (time-critical), since the same interface must serve both.

Out of scope

- Account registration, cancellation, TDR filing, refunds, and the post-booking experience.
- Ancillary products — flights, hotels, bus, tourism packages, e-catering — except where they compete for attention within the booking flow.
- Backend performance, availability and server capacity. These are real and widely reported, but they are engineering constraints rather than interface design decisions.
- Accessibility conformance testing (WCAG). Several findings have accessibility consequences, which are noted, but no formal audit was performed.

2. Why this product was selected

Three reasons made IRCTC a stronger choice than a conventional consumer app for this exercise.

2.1 Scale makes the cost of friction measurable

The platform handled between 1.33 crore and 1.74 crore logins on peak booking days while processing 17–20 lakh bookings. A five-second delay per booking, multiplied across that base, is a meaningful national cost in aggregate time. Few interfaces a student can evaluate carry that much leverage.

2.2 Users cannot exit, so the product does not self-correct

In a competitive market, a bad checkout loses customers and the business fixes it. IRCTC holds the inventory, so poor usability does not produce the feedback signal that normally forces improvement. Users respond instead by routing through third-party aggregators that sit on top of the same inventory — which transfers margin and relationship away from the operator without fixing the underlying experience. This is precisely the situation heuristic evaluation exists to address: the method does not require users to complain.

2.3 The user base is maximally diverse

The same screens must serve a first-time booker on a budget phone and a practised Tatkal user racing a thirty-minute inventory window. Designing for both is a genuine product problem, not a cosmetic one, and it exposes the tension between *simplicity* and *efficiency of use* that Nielsen's heuristics explicitly name as needing to coexist.

3. Method

The evaluation followed a standard expert heuristic review. The complete flow was walked twice — once adopting the mental model of a first-time traveller, once as an experienced Tatkal booker — and every screen was inspected against each of Nielsen's ten heuristics in turn. Candidate issues were then consolidated, de-duplicated, severity-rated, and cross-referenced against six UX laws to explain the underlying mechanism.

3.1 Severity rating scale

Rating	Meaning	Action implied
0	Not a usability problem	No action.
1	Cosmetic	Fix only if spare time is available.
2	Minor	Low priority; fix in normal backlog.
3	Major	High priority; important to fix.
4	Catastrophic	Imperative to fix before release. Users lose money, time or data.

3.2 Frameworks applied

Nielsen's ten usability heuristics: visibility of system status; match between system and the real world; user control and freedom; consistency and standards; error prevention; recognition rather than recall; flexibility and efficiency of use; aesthetic and minimalist design; help users recognise, diagnose and recover from errors; help and documentation.

UX laws applied: Hick's Law, Miller's Law, Jakob's Law, Fitts's Law, the Doherty Threshold, and the Aesthetic-Usability Effect. The heuristics answer *what* is broken; the laws explain *why it costs the user something* and how much.

3.3 Limitations of the method

- **No user testing was conducted.** Heuristic evaluation predicts problems; it does not measure them. Every severity rating in this report is an informed judgement, not an observed failure rate. The natural next step is a moderated test with five to eight participants split across the two personas.
- **No access to analytics.** Drop-off rates, error rates and time-on-task would sharpen prioritisation considerably and are unavailable externally.
- **Single evaluator.** A single reviewer typically surfaces roughly a third of the issues a panel of three to five would find. The nine issues here should be read as a floor, not a complete inventory.
- **Moving target.** A beta redesign launched in July 2026 and the flow may change further.

4. Users

Two personas were used to walk the flow. They are deliberately chosen to pull in opposite directions, because the central design tension in this product is that both must be served by one interface.

Kavya · 24 · first-time / infrequent

Context: Works in Bengaluru, travels home to Hubballi two or three times a year, mostly around festivals. Books on a mid-range Android phone over patchy 4G.

Knowledge: Does not know what RAC, GNWL, PQWL, Vikalp or "Reservation Choice" mean. Has heard of Tatkal but has never used it.

Goal: "Get a confirmed seat home for the holiday, and don't let me lose money by accident."

Biggest fear: Paying for a ticket that turns out not to be a real seat.

Rahul · 34 · frequent / Tatkal

Context: Travels monthly for work. Keeps a Master List of saved passengers and logs in five minutes before the window opens.

Knowledge: Fluent in the abbreviations. Knows AC Tatkal opens at 10:00 and non-AC at 11:00, one day before departure, and that seats can be gone inside a minute.

Goal: "Complete the entire flow in under sixty seconds without a single unnecessary keystroke."

Biggest fear: Losing the race to a captcha, an OTP round-trip or a mistyped field.

The design tension in one line

Kavya needs the interface to explain itself; Rahul needs it to get out of the way. The current design serves neither — it is simultaneously too cryptic for Kavya and too slow for Rahul. Progressive disclosure is the mechanism that resolves this, and it is largely absent.

5. The user flow, and where it stalls

#	Step	What the user is trying to do	Observed friction
1	Land and search	Enter origin, destination and date	The search widget shares the fold with a 12-item navigation bar, a scrolling alert marquee, six concession checkboxes, a promotional carousel, seven cross-sell tiles and an advertisement slot.
2	Compare trains	Decide which train and class to take	Twenty-eight filter chips at equal visual weight. Availability rendered as GNWL45/WL20, RAC 24, CURR_AVBL, REGRET — with the legend explaining them pushed below the fold.
3	Enter travellers	Supply names and ages	A seven-column table per passenger, six unexplained preference checkboxes, a "Reservation Choice" dropdown with four near-identical options, optional GST fields at the same visual weight as name.
4	Verify	Prove they are human and Aadhaar-verified	A distorted-text captcha followed by an Aadhaar OTP round-trip, both inside a countdown the user was not warned about.
5	Pay	Pay the amount quoted	Twenty-four payment tiles across thirteen categories, five of which are the same underlying rail (UPI), with fees expressed in five different formats.
6	Confirm	Know the booking succeeded	The amount debited can exceed the total quoted two screens earlier, because gateway charges are added on the bank page.

6. Screen-by-screen evidence

Numbered markers on each screen correspond to the annotations described beneath it. Issue numbers in **bold** refer to the consolidated findings in section 7.

The screenshot shows the IRCTC homepage. At the top is a navigation bar with links for IRCTC Tourism, IRCTC Flights, IRCTC Hotels, IRCTC Bus, Rail Drishti, Loyalty Programme, Tenders, and Ask Disha. On the right are links for Contact Us, Alerts, English, and a language selector (A- A+). Below this is a header section with the IRCTC logo, helpline number (14646), email (care@irctc.co.in), and booking hours (00:20 - 23:45 hrs). There are buttons for LOGIN, REGISTER, and AGENT LOGIN. A secondary navigation bar includes BOOK TICKET (highlighted with marker 1), TRAINS, IRCTC eWALLET, LOYALTY, MY TRANSACTIONS, CHARTS / VACANCY, TRACK YOUR TRAIN, e-CATERING, RETIRING ROOM, HOLIDAY PACKAGES, and SHUBH YATRA. Below this is a banner with alerts: Beware of fraudulent websites, Aadhaar authentication mandatory for Tatkal & ARP-opening bookings, Advance Reservation Period revised to 60 days, and Scheduled maintenance 23:45-00:20 hrs. The main content area is divided into two columns. The left column features a 'BOOK TICKET' section with a search form. The form has fields for 'From' (marker 6), 'To' (marker 2), and 'Date of Journey' (DD/MM/YYYY). Below these are dropdowns for 'Class' (set to 'All Classes') and 'Quota' (set to 'GENERAL'). A 'Search' button is to the right. Below the search form are six checkboxes for concessions: Person With Disability Concession, Flexible With Date, Train with Available Berth, Railway Pass Concession, Divyaang Concession, and Book Only If Confirm (marker 3). Below this is a promotional banner for 'Char Dham Yatra by Bharat Gaurav Train — Book Now!' (marker 4) with packages from ₹19,999 and EMI available. At the bottom of the left column is a horizontal carousel with buttons for Flights, Hotels, Bus, e-Catering, Loyalty, Retiring Room, and PNR Status. The right column features a sidebar with a 'IRCTC eWALLET — Faster Checkout' section (marker 5) showing one-time registration fee ₹50 and transaction charge ₹10 + GST. Below this is a 'QUICK LINKS' section with links for Cancel Ticket, File TDR, Booked History, Refund Status, Change Boarding Point, Train Schedule, Fare Enquiry, Seat Map, Vikalp (ATAS), Concession Rules, Divyaang ID, Reservation Rules, Tatkal Rules, and Baggage Rules. At the bottom of the sidebar is an 'Advertisement' slot (728 x 90 promotional slot) and a 'TRENDING OFFERS' section with a flat 10% off on IRCTC Air and zero convenience fee on UPI Lite.

Screen 1 — Homepage and search entry. ① A twelve-item primary navigation, most of which is unrelated to booking a train. ② Class and Quota default to values the user must understand to change, with no explanation of what a quota is. ③ Six concession checkboxes shown permanently, though they apply to a small minority of bookings. ④ A promotional carousel occupying prime space directly beneath the search widget. ⑤ A full sidebar of cross-sell, quick links and advertising competing with the primary task. ⑥ The one control that matters — origin station — has no more visual weight than anything around it. Relates to issues **1** and **4**.

IRCTC Tourism

IRCTC Flights

IRCTC Hotels

Rail Drishti

Ask Disha

Hi, RAHUL_S Logout English

BOOK TICKET

TRAINS

IRCTC eWALLET

MY TRANSACTIONS

CHARTS / VACANCY

TRACK YOUR TRAIN

MORE

From

To

Date

Class

Quota

Modify Search

KSR BENGALURU - SBC

MGR CHENNAI CTL - MAS

20/09/2026

All

GENERAL

28 Trains found from SBC to MAS on Sun, 20 Sep 2026

Sort: Departure | Show Available Only

AC

Non AC

1A

2A

3A

3E

SL

CC

EC

2S

FC

Rajdhani

Shatabdi

Duronto

Vande Bharat

Garib Rath

Superfast

Express

Mail

Tejas

Departure 00-06

06-12

12-18

18-24

Arrival 00-06

06-12

12-18

18-24

SBC

BNC

YPR

KJM

Advertisement

12658 — KSR BENGALURU MAS SF EXPRESS

SMTWTFSS

Train Schedule · Seat Availability · Coach Position

22:40

SBC | Sun, 20 Sep

— 5h 50m · 362 km —

04:30

MAS | Mon, 21 Sep

SL

AVAILABLE-0086

Free Cancellation · 91%

3A

RAC 24

76%

2A

GNWL45/WL20

31%

1A

REGRET/WL 112

—

3E

CURR_AVBL-0012

68%

Book Now

12610 — MYSURU MAS EXPRESS

SMTWTFSS

Train Schedule · Seat Availability · Coach Position

06:00

SBC | Sun, 20 Sep

— 7h 05m · 362 km —

13:05

MAS | Sun, 20 Sep

SL

PQWL18/WL 9

22%

3A

AVAILABLE-0004

88%

2A

RAC 3

71%

2S

AVAILABLE-0310

99%

CC

TQWL 6

—

Book Now

20607 — MAS VANDE BHARAT EXPRESS

SMTWTFSS

Train Schedule · Seat Availability · Coach Position

05:50

SBC | Sun, 20 Sep

— 4h 25m · 362 km —

10:15

MAS | Sun, 20 Sep

CC

AVAILABLE-0031

Dynamic fare applies · 94%

EC

AVAILABLE-0007

Dynamic fare applies · 89%

Book Now

Note: AVBL = Available · RAC = Reservation Against Cancellation · GNWL = General Waiting List · PQWL = Pooled Quota Waiting List · RLWL = Remote Location Waiting List · TQWL = Tatkal Waiting List · CURR_AVBL = Current Availability · REGRET = No further booking. Prediction % is indicative only and not guaranteed. Fares exclude convenience fee, catering and dynamic surcharge where applicable.

ALSO AVAILABLE

Flights SBC → MAA from ₹2,450

Bus SBC → Chennai from ₹699

Hotels near Chennai Central

TRAVEL INSURANCE

₹0.45 per passenger. Opt in on the next page.

Screen 2 — Search results. ① Twenty-eight filter chips presented as a single undifferentiated block, none collapsed or prioritised. ② Availability shown as GNWL45/WL20, RAC 24, CURR_AVBL-0012 and REGRET/WL 112, each paired with a confirmation percentage whose basis is not explained. ③ Fares are absent from the comparison entirely; "Dynamic fare applies" appears without a number. ④ The glossary that decodes every abbreviation above sits at the bottom of the page, after the decision has been made. Relates to issues 2, 3 and 7.

IRCTC Tourism

IRCTC Flights

Ask Disha

Hi, RAHUL_S

Logout

BOOK TICKET

TRAINS

IRCTC eWALLET

MY TRANSACTIONS

MORE

12658 — KSR BENGALURU MAS SF EXPRESS

SBC 22:40 → MAS 04:30 · Sleeper (SL) · GENERAL · Sun, 20 Sep 2026

Session expires in 02:11

4

Error: Please fill all mandatory details correctly and try again.

PASSENGER DETAILS

#	Name (as per Govt. ID)	Age	Gender	Berth Preference	Food Choice	Nationality
1	Rahul Sharma	34	Male	LB	Veg	India
2			Select	No Preference	No Food	India

+ Add Passenger

+ Add Infant (below 5 yrs, without berth)

Select from Master List

1

OTHER PREFERENCES

☐ Consider for Auto Upgradation

☐ Book only if confirm berths are allotted

☐ Avail Travel Insurance (₹0.45/passenger)

☐ Add Meal Preference (e-Catering)

☐ Apply Vikalp (ATAS) alternate train scheme

☐ I am a Divyaangjan / concession holder

Reservation Choice

Options: Book only if at least one lower berth is allotted · Book only if two lower berths are allotted · Book only if all passengers are allotted berths in same coach · Book if confirm berth

Coach ID (Optional)

Leave blank unless you know the coach layout.

Boarding Point

Changing this may affect your booking.

CONTACT & BILLING

Mobile Number

Email

GST Number (Optional)

Address for GST (Optional)

☐ Auto-upgrade my booking to a higher class if available at no extra cost (Terms apply)

3

VERIFICATION

x7Qm4b

Enter Captcha

Aadhaar OTP verification will be requested after this step.

2

Back

Continue

By continuing you accept the Terms of Service, Cancellation Policy, Refund Rules and Privacy Policy.

Screen 3 — Traveller details. ① Six preference checkboxes and a "Reservation Choice" dropdown whose four options differ only in fine detail, alongside an optional Coach ID field that assumes knowledge of coach layouts. ② The primary action sits beside a Back button styled as an equal peer. ③ A distorted-text captcha, with an Aadhaar OTP round-trip still to come. ④ A single generic error banner — "Please fill all mandatory details correctly" — that names neither the field nor the problem. ⑤ A silent 2:13 countdown in small red text, with no warning and no way to extend.

Relates to issues **4**, **5** and **6**.

IRCTC Tourism

Ask Disha

Hi, RAHUL_S

Logout

BOOK TICKET

TRAINS

IRCTC eWALLET

MY TRANSACTIONS

MORE

Payment

12658 · SBC → MAS · 20 Sep 2026 · Sleeper · 2 Passengers

IRCTC eWallet

BHIM / UPI / QR

Credit Card

Debit Card

Debit Card + ATM PIN

Net Banking

Wallets / Prepaid Cards

Multiple Payment Service

Pay on Delivery

EMI / Pay Later

Bharat QR

Rewards / Loyalty Points

International Cards

IRCTC eWallet

₹10 + GST

Paytm UPI

No fee

PhonePe

No fee

Google Pay

No fee

BHIM UPI

No fee

Visa / Master CC

₹23.60 + 1.8% MDR

RuPay CC

₹23.60

Amex

₹23.60 + 2.4% MDR

Debit Card < ₹2000

₹11.80

Debit Card ≥ ₹2000

₹23.60 + 0.9%

SBI Net Banking

₹11.80

HDFC Net Banking

₹11.80

ICICI Net Banking

₹11.80

Axis Net Banking

₹11.80

PNB Net Banking

₹11.80

Other Banks (48)

₹11.80

Amazon Pay

₹11.80

Mobikwik

₹11.80

Freecharge

₹11.80

PayPal (Intl.)

3.5% + GST

Pay on Delivery

₹? see T&C

EMI (6/9/12 mo)

Interest as per bank

Bharat QR

No fee

Loyalty Points

Redeem ≥ 500 pts

AMOUNT PAYABLE

Ticket Fare

₹ 1,050.00

Convenience Fee

₹ 23.60

Insurance

₹ 0.90

PG Charges

To be added

Payable Now

₹ 1,074.50*

*Excludes gateway charges.

Advertisement

Back

Pay & Book Ticket

Important: The convenience fee shown above is indicative. Payment gateway charges levied by the bank are additional and will be displayed on the bank page. IRCTC is not responsible for charges levied by third-party gateways. In case of payment failure with amount debited, refund will be processed within 5-7 working days. Do not press Back or Refresh.

Complete payment within 09:55

Screen 4 — Payment. ① Twenty-four tiles across thirteen left-hand categories; UPI appears five separate times under different names. ② The primary action again sits beside an equally weighted Back button, on a page that simultaneously warns against pressing Back. ③ Fine print stating that gateway charges are additional and that IRCTC is not responsible for them. ④ A total marked with an asterisk and the line "PG Charges — to be added", meaning the figure shown is explicitly not the figure that will be debited. ⑤ A ten-minute countdown introduced at the moment the user most needs to read carefully. Relates to issues 7, 8 and 9.

IRCTC booking flow · heuristic evaluation · September 2026

9

7. Consolidated findings

#	Issue	Heuristic violated	UX law	Severity
1	Homepage buries the primary task	Aesthetic & minimalist design; recognition rather than recall	Hick's Law	3
2	Availability shown in back-office notation	Match between system and the real world	Miller's; Jakob's	4
3	Twenty-eight undifferentiated filter chips	Aesthetic & minimalist design; flexibility & efficiency of use	Hick's Law	2
4	Traveller form asks for everything at once	Aesthetic & minimalist design; recognition rather than recall	Miller's; Hick's	3
5	Errors generic, late and detached from the field	Help users recognise, diagnose & recover; error prevention	Doherty	3
6	Session expires silently and destroys work	Visibility of system status; user control & freedom; error prevention	Doherty	4
7	Total changes after the user has committed	Visibility of system status; consistency & standards	Jakob's Law	4
8	Twenty-four payment tiles, five of them duplicates	Consistency & standards; aesthetic & minimalist design	Hick's; Fitts's	3
9	Going back is punished rather than supported	User control & freedom; error prevention	Jakob's Law	3

7.1 Detailed findings

ISSUE 1 Homepage buries the one task most visitors came to do

Severity 3

Evidence

The Book Ticket widget shares the first screen with a twelve-item navigation bar, a scrolling alert marquee, six concession checkboxes, a promotional carousel, seven cross-sell tiles and an advertising slot.

Why it matters

The overwhelming majority of traffic arrives to do exactly one thing. Every competing element extracts attention from a task the user had already decided on before arriving.

Heuristic

Aesthetic and minimalist design; recognition rather than recall.

UX law

Hick's Law — decision time rises with the number and complexity of choices presented.

Recommend

Give the search widget the entire first screen. Move Flights, Hotels, Bus and Tourism into a secondary tab. Collapse the alert marquee into a single dismissible line. Hide the six concession checkboxes behind an "Add a concession" control.

ISSUE 2 Availability is written in railway back-office code**Severity 4**

Evidence	Class tiles read GNWL45/WL20, RAC 24, CURR_AVBL-0012 and REGRET/WL 112. The glossary decoding these terms is positioned below the train list, after the decision point.
Why it matters	This is the single most consequential decision in the flow — whether the user will actually have a seat — and it is expressed in a notation that exists for railway operations, not for travellers. Kavya cannot tell whether RAC means she is travelling or not. A user who cannot distinguish a confirmed berth from a waiting list can spend money on a journey they cannot take.
Heuristic	Match between system and the real world.
UX law	Miller's Law (each tile carries three unexplained data points); Jakob's Law (no comparable booking product requires users to learn a code).
Recommend	Lead with plain language: "45 people ahead of you — bookings like this confirm about 31% of the time." Retain the abbreviation as secondary text so Rahul loses nothing. Make every status tappable to reveal what happens if it does not confirm, and what the refund position is.

ISSUE 3 Twenty-eight filter chips, all at equal visual weight**Severity 2**

Evidence	AC/non-AC, nine class codes, eight train types, eight time buckets and four origin stations are presented as one continuous block of chips, none collapsed, none prioritised, with results sorted by raw departure time.
Why it matters	Filters are a tool for reducing choice, so presenting them as an unstructured mass of choices is self-defeating. Rated 2 rather than higher because the chips are ignorable — a user can scroll past them and still complete the task.
Heuristic	Aesthetic and minimalist design; flexibility and efficiency of use.
UX law	Hick's Law.
Recommend	Show four high-intent filters — seats available, departure window, journey duration, AC — and place the remainder behind "More filters". Sort by best fit rather than departure order. Remember each user's habitual filters between sessions.

ISSUE 4 The traveller form asks for everything at once

Severity 3

Evidence	A seven-column table per passenger (name, age, gender, berth preference, food choice, nationality), plus six preference checkboxes, a "Reservation Choice" dropdown with four near-identical options, an optional Coach ID field, and optional GST fields — all at the same level of visual priority as name and age.
Why it matters	Only three fields are genuinely required to book. Everything else is optional but indistinguishable from mandatory, so cautious users like Kavya read all of it, and fast users like Rahul must visually skip past it under time pressure. The "Reservation Choice" options are the clearest example: four long sentences that differ by a few words each, presented in a dropdown.
Heuristic	Aesthetic and minimalist design; recognition rather than recall.
UX law	Miller's Law; Hick's Law.
Recommend	Ask for name, age and gender only. Collapse berth and meal preferences behind an optional disclosure, with an explicit note that skipping them does not reduce the chance of getting a seat. Put GST behind "I need a GST invoice". Replace the "Reservation Choice" dropdown with a plain question — "If seats don't confirm, what should we do?" — offering three clearly distinct answers.

ISSUE 5 Errors are generic, late, and detached from the field

Severity 3

Evidence	Submitting an incomplete form produces one banner at the top of the page reading "Please fill all mandatory details correctly and try again." Offending fields receive a red outline but no message explaining what is wrong.
Why it matters	The message names neither the field nor the fault, and arrives only after submission — the most expensive possible moment inside a countdown. A red outline with no text is also invisible to screen-reader users and to anyone with a red-green colour vision deficiency, so the only error signal present fails for part of the user base entirely.
Heuristic	Help users recognise, diagnose and recover from errors; error prevention.
UX law	Doherty Threshold — feedback that arrives late breaks the user's flow state.
Recommend	Validate on blur rather than on submit. Place a specific message beneath each field — "Add an age to continue" — rather than a generic banner. On submit, scroll to and focus the first error. Never clear data the user has already entered.

ISSUE 6 The session timer is a silent deadline that destroys work**Severity 4**

Evidence	A countdown appears in small red text in the page header. There is no warning before expiry, no way to extend, and expiry returns the user to search with entered passenger data lost.
Why it matters	This converts a recoverable delay into total data loss, and it does so in the flow where users are most time-pressured. For a Tatkal booker, expiry does not merely cost a retype — the inventory is gone by the time they return. The timer is also framed as a threat rather than as what it actually is: a seat being held for them.
Heuristic	Visibility of system status; user control and freedom; error prevention.
UX law	Doherty Threshold.
Recommend	Reframe the countdown as reassurance — "Seats held for 9:42" — rather than as a threat. Warn at two minutes with a one-tap Extend. Persist the form so that expiry never costs a retype. Remove the captcha entirely for Aadhaar-verified accounts, since the account has already been authenticated.

ISSUE 7 The total changes after the user has committed**Severity 4**

Evidence	The fare summary quotes a total, then marks it with an asterisk: payment gateway charges are shown as "to be added" and are only revealed on the bank's page. Each of the twenty-four payment tiles carries a different fee, expressed inconsistently.
Why it matters	A price that changes after commitment is the single most trust-damaging pattern in commerce. It is severity 4 because it costs users money they did not agree to, and because it happens at the exact moment they are least able to reconsider.
Heuristic	Visibility of system status; consistency and standards.
UX law	Jakob's Law — every comparable travel product quotes an all-inclusive price, so users reasonably expect this one to.
Recommend	Quote one all-inclusive total from the results page through to the receipt. Where a payment method genuinely costs more, show the difference on the tile before it is selected. Default to the zero-fee option.

ISSUE 8 Twenty-four payment tiles, five of which are the same thing**Severity 3**

Evidence	Thirteen left-hand categories and twenty-four tiles. UPI appears as BHIM/UPI, Paytm UPI, PhonePe, Google Pay and Bharat QR. Fees are written as "₹11.80", "₹23.60 + 1.8% MDR", "3.5% + GST", "Interest as per bank" and "₹? see T&C".
Why it matters	The user must resolve which of five superficially different options is the same underlying rail, and then compare five incommensurable fee formats — all inside a ten-minute countdown. The dense four-across grid also produces small targets on mobile.
Heuristic	Consistency and standards; aesthetic and minimalist design.
UX law	Hick's Law; Fitts's Law.
Recommend	Surface the user's last-used method first, followed by three defaults — UPI, card, netbanking — with everything else behind "Other ways to pay". Use one fee format throughout. Collapse the five UPI entries into a single UPI option that then offers the app choice.

ISSUE 9**Going back is punished rather than supported****Severity 3****Evidence**

The payment page instructs the user not to press Back or Refresh. Returning from payment loses the selection, and there is no visible way to edit traveller details without restarting the flow.

Why it matters

Users change their minds; this is normal behaviour, not misuse. Instructing people not to use a standard browser control shifts responsibility for a system limitation onto the user, and the warning appears precisely when second-guessing is most likely.

Heuristic

User control and freedom; error prevention.

UX law

Jakob's Law.

Recommend

Remove the warning by making Back safe. Hold the seat server-side so returning costs nothing. Make every completed step editable from a persistent trip summary rail.

8. Mapping to UX laws

Nielsen's heuristics identify what is broken. The laws below explain the mechanism — why each defect imposes a measurable cost on the user rather than merely being untidy.

Law	Where it bites in this flow	What to do about it
Hick's Law Decision time rises with the number and complexity of choices.	Twenty-eight filter chips; twenty-four payment tiles across thirteen categories; six concession checkboxes on the homepage. A Tatkal user facing a thirty-minute inventory window is asked to make at least five non-essential decisions along the way.	Cap visible options at three to five per decision point and defer the remainder behind progressive disclosure. Rank by likely intent rather than presenting alphabetically or by internal category.
Miller's Law Working memory holds roughly seven $\pm$ two chunks.	A seven-column passenger table repeated for each traveller; five class tiles each carrying a code, a count and a percentage. The user exceeds working-memory capacity before entering a single name.	Chunk the form into "who is travelling" and then "preferences". Reduce each class tile to one plain-language status and one all-inclusive price.
Jakob's Law Users expect a product to behave like the others they already use.	Saved travellers, all-inclusive pricing, one-tap UPI and an editable summary are standard across the travel apps Indians use daily, and are largely absent here — which is a substantial part of why many book rail through aggregators instead of direct.	Adopt the established booking pattern rather than inventing a rail-specific one. Novelty in a checkout flow is a cost, not a feature.
Fitts's Law Time to acquire a target depends on its distance and size.	Roughly eleven-pixel checkboxes; a dense four-across payment grid; a primary "Book Now" sitting beside three text links of similar prominence, on screens frequently operated one-handed at speed.	Enforce a 44 × 44 pixel minimum target. Pin the primary action within thumb reach on mobile, and separate it visually from adjacent destructive or navigational controls.
Doherty Threshold Attention and flow hold when the system responds within about 400 ms.	A captcha, then an Aadhaar OTP round-trip, then a gateway redirect — all inside a countdown, with no optimistic feedback and no skeleton states. Perceived slowness during Tatkal converts directly into abandoned bookings.	Add skeleton and optimistic states, prefetch the next step, and remove verification steps that a verified account has already satisfied.
Aesthetic-Usability Effect People perceive attractive interfaces as easier to use, and forgive their flaws more readily.	A dated visual language lowers perceived trustworthiness at the precise moment money changes hands — on the one platform that is actually the authoritative source of rail inventory.	Move to a calm, modern visual system. IRCTC's own July 2026 beta interface already points in this direction; the opportunity is to pair the visual refresh with the structural changes above rather than shipping it alone.

9. Prioritisation

Severity alone does not determine sequence; implementation cost does too. Plotting the nine issues against impact and effort produces a clear first release.

	Low effort	High effort
High impact	Do first 2 · Translate the jargon7 · One all-in price 5 · Inline errors6 · Seat hold + extend Mostly copy, validation and default changes. No new architecture.	Plan next 4 · Progressive disclosure9 · Editable summary 8 · Ranked payment methods Requires state management, server-side seat holds and gateway work.
Lower impact	Fill-in work 3 · Collapse filters1 · Declutter homepage Worth doing, but users can work around both today.	Nothing sits here — which is a good sign that the backlog is not carrying expensive low-value work.

Why these four go first

Issues 2, 5, 6 and 7 account for all three severity-4 findings plus the most common source of wasted effort. Critically, none of them requires re-architecting the booking engine: translating availability into plain language is a content and presentation change; quoting an all-inclusive price is an arithmetic change; inline validation is a front-end change; and persisting form state across session expiry is a well-understood pattern. This is an unusually favourable ratio of user benefit to engineering cost.

10. Redesign concept

The following screens illustrate the recommendations applied. They are a concept intended to make the principles concrete, not a finished specification. Green markers indicate what changed.

IRCTC Book a train 1 Choose train 2 Travellers 3 Pay Rahul S.

Bengaluru (all stations) → Chennai (all stations) Sun, 20 Sep 2026 · 2 travellers Change

2 Seats available Morning departure Under 5 hours AC only More filters 9 of 28 trains match · Sorted by best fit

3

Vande Bharat Express · 20607
Runs daily · Meals included

05:50 Bengaluru SBC 4h 25m direct 10:15 Chennai MAS

Chair Car
31 seats left
₹1,240 total, all fees in

Executive Chair
7 seats left
₹2,390 total, all fees in

Select

4

Bengaluru-Chennai SF Express · 12658
Runs daily · Overnight

22:40 Bengaluru SBC 5h 50m arrives next day 04:30 Chennai MAS

Sleeper · no AC
86 berths left
₹560 total, all fees in

AC 3-Tier
You will travel · shared seat
Becomes a full berth if 24 people cancel — happens 76% of the time

AC 2-Tier
Waiting list · 45 people ahead
Only 31% of bookings like this confirm. You can choose a backup below.

Select

5

Plain-language glossary is now built in. Hovering or tapping any status shows what it means and what happens if it does not confirm — no railway abbreviations are shown to first-time travellers. Fares are quoted all-inclusive, so the number never changes at payment.

Choosing a train — after. ① A three-step progress indicator establishes how much of the flow remains, and the search summary stays editable in place. ② Four intent-based filters replace twenty-eight code-based chips; the rest move behind "More filters", and results are sorted by fit. ③ Availability is expressed in plain language — "31 seats left" — alongside an all-inclusive fare, so the comparison the user is actually making is the comparison the screen supports. ④ Waiting-list and RAC states are translated into consequences: what will happen to you, and how often that outcome occurs. The abbreviation is retained as secondary text for experienced users. ⑤ The glossary is built into the interface at the point of need rather than parked at the foot of the page.

Addresses issues 2, 3 and 7.

Book a train Choose train **Travellers** Pay Seats held for 9:42 · Extend

Who is travelling?

Rahul Sharma · 34 · Male · saved + Add traveller

Full name, as printed on the ID you will carry Age Add an age to continue Gender Female

Priya Sharma

Berth & meal preferences (optional) Collapsed by default. Skipping this does not reduce your chance of getting a seat.

If seats do not confirm

Refund my money
Book only if every traveller gets a confirmed berth. Full refund otherwise.

Put me on any similar train
We move you to another train on the same route at no extra cost.

Keep me on the waiting list
You will travel only if it confirms. We will text you the moment it does.

Where you board

KSR Bengaluru · 22:40 · nearest to you Bengaluru Cantt. · 22:58 Krishnarajapuram · 23:14

Back to trains Continue to payment Nothing is charged yet. Your choices are saved if you go back.

Your trip

Vande Bharat Express · Chair Car
Bengaluru 05:50 → Chennai 10:15
Sun, 20 Sep 2026 · 2 travellers

Seats (2) ₹2,400
Booking & payment fees ₹80

You pay **₹2,480**
This is the final amount. It will not change at payment.

Free cancellation until 48 hours before departure. Cancel in two taps from "My trips".

Entering travellers — after. ① Saved travellers are offered first, so the common case is one tap rather than six fields. ② Berth and meal preferences are collapsed by default, with an explicit reassurance that skipping them does not reduce the chance of getting a seat — removing the anxiety that currently makes cautious users read everything. ③ The opaque "Reservation Choice" dropdown becomes a plain question with three clearly distinct answers, stated as outcomes rather than as policy language. ④ A persistent trip summary carries the final, all-inclusive amount and states explicitly that it will not change at payment. ⑤ The countdown is reframed as a seat hold the user controls, with an Extend action rather than a silent expiry. Inline field errors name the specific problem. Addresses issues 4, 5, 6, 7 and 9.

11. How we would know it worked

Recommendations without measurement are opinions. Each proposed change maps to an observable metric.

Metric	Why it is the right measure	Issues it tests
Median search-to-payment time , segmented for General and Tatkal	Directly captures the Hick's and Miller's Law reductions. Segmenting matters because the two personas have opposite definitions of success.	1, 3, 4, 8
Drop-off rate at the traveller and payment steps	These are the two steps carrying the heaviest cognitive load and the countdown; they should move first if the changes work.	4, 6, 8
Form error rate per completed booking	Isolates the effect of inline validation from everything else.	5
Session-expiry restarts per booking	A direct count of destroyed work; should approach zero once state persists.	6
Support contacts about unexpected charges	The most honest signal on pricing transparency, and already an operational cost centre.	7
Share of bookings made direct versus via aggregators	A slow-moving but strategically important indicator of whether the official channel is winning back users on experience.	All
Comprehension test on availability states	Ask first-time users what a given status means before and after. The most direct test of the highest-severity finding.	2

12. Conclusion

The nine issues documented here are not the result of carelessness. They are what happens when an interface grows outward from an operational system: each addition was locally rational — another quota, another payment option, another concession checkbox — and the cumulative effect is an interface that asks the traveller to learn the railway's vocabulary in order to buy a ticket.

The most valuable finding is also the most tractable. Three of the four highest-priority fixes are translation and transparency changes rather than architectural ones: say what the availability actually means, quote the price you will actually charge, and tell the user specifically what went wrong. None of these requires rebuilding the booking engine, and together they address all three severity-4 findings.

The appropriate next step is validation. A moderated usability test with five to eight participants, split between first-time and frequent bookers, would convert the severity estimates in this report into observed failure rates — and would almost certainly surface issues a single evaluator missed.

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A note on the screens in this report

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